# Development and Industrial Implementation of Pakistan's First Integrated Biofuel Production Process: A Pathway Toward Green Energy Transition

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## 1 Abstract

This study presents Pakistan's first successful industrial implementation of bio-jet fuel production, marking a transformative milestone in the nation's transition toward sustainable energy. Developed and operated by Bio Tech Energy (Pvt.) Ltd. under SAFCO Ventures, the project establishes the country's first commercial-scale biodiesel and bio-jet fuel facility—an unprecedented achievement in South Asia's emerging green energy landscape. Utilizing locally available waste-based feedstocks such as used cooking oil, poultry fat, and other agricultural by-products, the process converts high-FFA organic waste into high-purity fatty acid methyl esters (FAME) through optimized pre-esterification and trans-esterification reactions, followed by advanced distillation and methanol recovery systems. The plant's integrated design demonstrates high resource efficiency, circular utilization of industrial by-products, and near-zero liquid discharge, thereby minimizing environmental footprint. Beyond its technical innovation, this initiative leverages Pakistan's agricultural strength to create a replicable bio-refining model that supports rural value chains, waste management, and low-carbon aviation fuel production. The project not only establishes a foundation for large-scale Sustainable Aviation Fuel (SAF) development but also aligns with the global "dual-carbon" and energy-transition goals. This pioneering step positions Pakistan as an emerging contributor to the international clean-energy ecosystem and demonstrates how developing economies can transform waste into strategic energy resources through industrial innovation and sustainability-driven engineering.

**Keywords:** Sustainable Aviation Fuel (SAF); Waste-to-Energy Conversion; Bio-Jet Fuel Production; Circular Bioeconomy; Used Cooking Oil (UCO) Valorization; Low-Carbon Energy Transition; Industrial Biofuel Implementation

## 2. Introduction

Pakistan ranks among the most climate-vulnerable countries globally, having suffered catastrophic events such as the 2022 floods that displaced over eight million people, caused over US$30 billion in damage, and underscored the urgency for deep decarbonization and resilient energy systems [1], [2]. Simultaneously, Pakistan produces large volumes of organic and waste oils from urban food services, agriculture, and industrial streams—used cooking oil alone in Punjab is estimated at hundreds of thousands of tonnes per annum—representing an under-utilized resource for biofuel production [3]. Despite this potential, policy frameworks, regulatory incentives, and investment in the biofuels and SAF sectors have remained nascent, with limited institutional focus until recent announcements, such as the International Finance Corporation's US$35 million financing and the Asian Development Bank's technical assistance program, to establish Pakistan's first large-scale SAF facility in Sheikhupura, Punjab [4], [5]. This paper presents **Bio Tech Energy's** pioneering industrial-scale bio-jet fuel project—Pakistan's first such undertaking—to bridge this gap, demonstrating both technical feasibility and strategic value in harnessing municipal and agricultural residues for low-carbon aviation fuel as part of the national green-energy transition.

While Pakistan's exposure to climate extremes has been repeatedly highlighted by global assessments, the domestic waste landscape presents an equally urgent environmental and developmental challenge. The country generates approximately 49–50 million metric tonnes of solid waste annually, growing at about 2.4 % per year, with more than half comprising organic matter suitable for conversion into energy resources [6], [7]. In major urban centres such as Karachi and Lahore, daily municipal solid waste (MSW) generation exceeds 16,000 t/day and 7,600 t/day, respectively [8]. The average organic content of MSW is reported at around 58 %, with food and yard residues as dominant fractions [7]. These data, summarized in Table 1, indicate a vast, underutilized feedstock reservoir for biofuel production that aligns directly with the objectives of SAF development. Despite this potential, Pakistan's policy and institutional framework for waste valorization and biofuel development remains at a formative stage. Recent instruments such as the National Hazardous Waste Management Policy (2022) and provincial waste regulations emphasize segregation and recycling but lack specific mechanisms for large-scale bioenergy or SAF integration [9], [10]. Reports from the CoRe Alliance and UNFCCC Biennial Transparency Report (2024) further underscore structural deficiencies in enforcement, inter-agency coordination, and incentive structures [11], [12]. Consequently, the mismatch between resource potential and regulatory preparedness highlights a critical vacuum in Pakistan's green-energy policy architecture. It is precisely within this void that Bio Tech Energy (Pvt.) Ltd., under SAFCO Ventures, has positioned its industrial-scale bio-jet fuel project—the nation's first initiative to transform organic and agricultural residues into certified low-carbon aviation fuel, thereby bridging the long-standing gap between theoretical potential and operational reality.

Situated within a dense agro-industrial corridor, Sheikhupura benefits from proximity to Lahore, Gujranwala, Sialkot, and Faisalabad — urban/industrial catchments that generate significant volumes of used cooking oil (UCO), animal fats, and agricultural residues—making it a strategic node for scalable waste-to-fuel collection; Pakistan's first SAF facility is sited in Sheikhupura and is backed by landmark commitments from IFC (up to US$35 million) and ADB (US$86.2 million), underscoring feedstock visibility and bankability [13] – [15]. At the same time, empirical studies highlight logistics frictions typical of emerging UCO markets—fragmented, informal collection networks and variable fuel quality — observable in Pakistan and specifically documented in Lahore, necessitating robust pre-treatment, traceability, and supplier formalization within the supply chain [16], [17]. On the demand side, Pakistan's aviation market is materially integrated with international traffic and is expected to keep expanding, while external mandates (e.g., EU ReFuelEU Aviation's phased SAF blending requirements) and airline decarbonization targets create export-aligned pull for certified waste-based SAF [18], [20]. Yet policy analyses consistently diagnose regulatory and institutional gaps — limited enforcement, incentive misalignment, and under-specified bioenergy/SAF pathways—which have historically constrained uptake despite resource potential [19]. Against this backdrop, our contribution is to present the first industrial-scale Pakistani initiative — Bio Tech Energy / SAFCO's waste-oil-to-bio-jet implementation — detailing the process architecture (feedstock pre-treatment, optimized trans-esterification, integrated distillation and methanol recovery) and early performance indicators, and to frame its socio-economic relevance (local value-chains, employment) and replicability.

The remainder of this paper is structured as follows: Section 2 presents the Materials and Methodology, detailing the industrial process configuration, operational stages, and analytical controls employed throughout the production chain. Section 3 discusses the Results and Applications, including process performance, efficiency metrics, and compliance with international fuel-quality standards. Finally, Section 4 provides the Conclusions, emphasizing the technological, environmental, and national implications of this initiative in advancing Pakistan's renewable-energy transition, with claims supported by publicly available disclosures where internal data remain confidential [13] – [15], [18] – [20].

**Table 1:** Key Studies and Sources on Pakistan's Waste Generation and Policy Landscape

| Ref. No. | Study / Report | Key Data / Findings | Relevance to Bio-Waste & Policy Gaps |
|---|---|---|---|
| [6] | *Pakistan's 49.6 Million Tonnes Solid Waste Potential – Source of Energy Recycling for Circular Economy*, SDPI / Pakistan Today (2024). | ≈ 49.6 Mt solid waste per year; low segregation; unrealized energy potential. | Establishes scale of SWM; identifies circular-economy opportunity. |
| [7] | Korai et al., *Cooperative Environmental Governance in Urban South Asia: Waste Management and Energy Potential in Pakistan*, PMC (2023). | ≈ 30.8 Mt MSW annually; ≈ 58 % organic fraction; bioenergy potential via anaerobic digestion and pyrolysis. | Quantifies organic fraction and theoretical energy yield. |
| [8] | *Country Commercial Guide: Waste Management in Pakistan*, U.S. Department of Commerce (2023). | Karachi ~16,500 t/day; Lahore ~7,690 t/day; food waste 30 %, yard waste 14 %. | Provides feedstock distribution for urban biofuel logistics. |
| [9] | *National Hazardous Waste Management Policy 2022*, Ministry of Climate Change, Govt. of Pakistan. | Introduces national guidelines for waste segregation and hazard control. | Framework document but no explicit bioenergy mandate. |
| [10] | *Policy Sought to Regulate Waste Management*, DAWN News (2025). | CoRe Alliance calls for recycling and fuel conversion policies. | Illustrates emerging advocacy and policy gaps. |

| Ref. No. | Study / Report | Key Data / Findings | Relevance to Bio-Waste & Policy Gaps |
|---|---|---|---|
| [11] | *Pakistan Biennial Transparency Report to UNFCCC* (2024). | ~ 0.65 kg/capita/day MSW rate; open burning and methane emissions quantified. | Links waste mismanagement to climate emissions. |
| [12] | *Trade.gov Market Assessment – Waste Management Sector Pakistan* (2024). | Forecasts > 2.4 % annual waste growth; need for PPP investments in energy recovery. | Demonstrates institutional interest but limited execution. |

# Materials and Methodology

The following section outlines the industrial process framework, equipment configuration, and operational methodology employed in the production of biodiesel and bio-jet fuel at Pakistan's first integrated biofuel facility. Each subsection systematically describes a specific stage of the process—from **feedstock preparation and moisture control** to **esterification, separation, purification, and solvent recovery**—to illustrate the complete conversion of waste-based oils into high-purity renewable fuel within a closed-loop system.

## 2.1 Raw Material Receiving, Settling and Impurity Removal

The process commences with the structured receipt and conditioning of heterogeneous feedstocks comprising material delineated in Table 2. Upon delivery, feedstocks are off-loaded into a **dedicated raw-material storage area** consisting of heated, bunded tanks fitted with level gauges, temperature sensors, and spill-containment systems to ensure traceability and environmental compliance. From these storage units, the oils are transferred through pre-heating lines to a **tri-tank settling system** designed to stabilize and homogenize the raw input stream before chemical processing. Within the settling tanks, controlled heating at approximately 80–90 °C reduces viscosity and expedites phase separation. Over an 8–10 hour residence period, free water and dense impurities gravitate to the tank bottoms, while light fractions form a scum layer at the top. Periodic skimming removes surface contaminants, and bottom drains discharge aqueous and sludge residues to a wastewater collection network for subsequent treatment. **Figure 1** shows the some of the most common raw materials procured and processed.

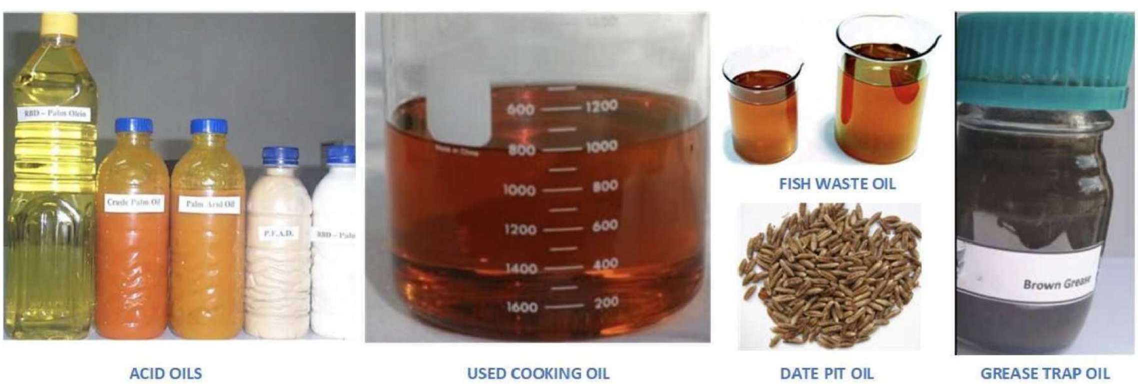

**Figure 1: Raw materials procurement lot**

The partially clarified oil then undergoes sequential **chemical and mechanical polishing**. A mild caustic or phosphoric-acid wash eliminates residual gums and phospholipids, followed by a brine rinse to neutralize soaps and stabilize the oil. The product stream passes through a **plate-and-frame filter-press assembly**, capturing fine particulates and yielding an optically clear feed with moisture typically below 1 % w/w. Laboratory quality-control analyses verify parameters such as acid value, sediment content, and residual moisture before transfer to the dehydration system. This preliminary purification phase forms the cornerstone of process fidelity, ensuring consistent feedstock quality for the downstream catalytic steps. Excessive impurities or entrained water would otherwise induce emulsification and catalyst deactivation during esterification. By achieving a uniform, contaminant-free feedstock, the plant secures stable reaction kinetics and prolonged catalyst life, directly influencing conversion efficiency and overall yield performance. Storage area is shown in **Figure 2**.

**Table 2: Primary Raw Materials for Biodiesel Production and Their Typical Chemical Structures**

| Feedstock Source | Typical Major Fatty Acids / Esters Formed (FAME) | Representative Chemical / Ester Formula | General Characteristics |
|---|---|---|---|
| Poultry Feather Acid Oil | Oleic ($C_{18:1}$), Palmitic ($C_{16:0}$), Linoleic ($C_{18:2}$) | Predominant methyl oleate → $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | High free-fatty-acid content; requires pre-esterification. |
| Palm Oil | Palmitic ($C_{16:0}$), Oleic ($C_{18:1}$), Linoleic ($C_{18:2}$) | $CH_3(CH_2)_{14}COOCH_3$ / $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | Semi-solid at room temperature; good oxidative stability. |
| Soybean (Soya) Acid Oil | Linoleic ($C_{18:2}$), Oleic ($C_{18:1}$), Palmitic ($C_{16:0}$) | $CH_3(CH_2)_4CH=CHCH_2CH=CH(CH_2)_7COOCH_3$ | Unsaturated; high iodine number; good cold-flow properties. |
| Rice Bran Oil | Oleic ($C_{18:1}$), Linoleic ($C_{18:2}$), Palmitic ($C_{16:0}$) | $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | Moderate FFA; rich in γ-oryzanol; yields good lubricity. |
| Used Cooking Oil (UCO) | Variable – mainly Oleic ($C_{18:1}$), Linoleic ($C_{18:2}$), Palmitic ($C_{16:0}$) | Mixed methyl esters; dominant $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | High FFA; water and polymerized residue require pre-treatment. |
| Cottonseed Oil | Linoleic ($C_{18:2}$), Palmitic ($C_{16:0}$), Oleic ($C_{18:1}$) | $CH_3(CH_2)_4CH=CHCH_2CH=CH(CH_2)_7COOCH_3$ | Good cold-flow; potential gossypol impurities need removal. |
| Food-Waste Oil | Broad mix – Oleic ($C_{18:1}$), Palmitic ($C_{16:0}$), Stearic ($C_{18:0}$) | $CH_3(CH_2)_{16}COOCH_3$ / $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | Highly variable; composition depends on feed origin and frying cycles. |
| Canola Oil | Oleic ($C_{18:1}$), Linoleic ($C_{18:2}$), Linolenic ($C_{18:3}$) | $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ / $CH_3(CH_2)CH=CHCH_2CH=CH(CH_2)_7COOCH_3$ | Low-sulfur, high-monounsaturated; excellent oxidative stability. |
| Palm Oil (duplicate feedstock, export grade) | Same as above | $CH_3(CH_2)_{14}COOCH_3$ / $CH_3(CH_2)_7CH=CH(CH_2)_7COOCH_3$ | Frequently blended with UCO for consistent viscosity. |

**Figure 2: Feed-stock Receiving and Settling Storage Tank Area**

## 2.2 Moisture Removal (Dehydration)

As a continuation of the feedstock pre-treatment sequence initiated at the storage and settling stage, the dehydration process serves to eliminate residual or chemically bound moisture that persists after gravity separation and washing. While the preceding phase ensures removal of free water and gross impurities, trace moisture—typically below 1 % w/w—can still induce catalyst hydrolysis and soap formation during esterification. Hence, a dedicated **vacuum-drying operation** is employed to achieve the stringent dryness required for stable reaction kinetics; see **Figure 3 (a)**.

Clarified oil from the settling system is transferred to dual **dehydration tanks (D101 A/B)**, each equipped with internal heating coils and connected to a falling-film evaporator (E101) driven by heater E201. Under a controlled vacuum of –0.3 to –0.5 bar, the oil is heated to approximately 105–110 °C, allowing entrained and dissolved water to vaporize without thermal degradation of the feed. Vapor condensate is collected in a closed receiver and discharged to wastewater treatment, while dehydrated oil is cooled to 60–70 °C prior to transfer into the pre-esterification reactors; **Figure 3 (b)** show the GST plant. Continuous sampling and laboratory verification confirm a final moisture content of ≤ 0.3 %, ensuring compatibility with acid-catalyzed reaction systems. Operational synchronization between the heating, vacuum, and condensate-recovery loops minimizes thermal losses and prevents oxidative deterioration. This controlled dehydration step thus completes the feedstock conditioning phase—establishing the necessary physicochemical congruence between the raw-oil characteristics and the process requirements for efficient esterification and trans-esterification downstream.


**Figure 3: (a) D101A and D101B vessels for removal of moisture + GST Tank (b) GST AGITATOR REACTOR**

## 2.3 Pre-Esterification (Free Fatty Acid Reduction)

Following dehydration, the conditioned feedstock—now free from suspended solids and residual moisture—undergoes acid-catalyzed **pre-esterification** to suppress the **free-fatty-acid (FFA)** content that would otherwise neutralize the alkaline catalyst in subsequent **trans-esterification**. The reaction **(Figure 4)** converts carboxylic acids present in the oil into **FAME** through reversible esterification with methanol, liberating water as a by-product:

$R–COOH + R_1OH \rightleftharpoons R–COOR_1 + H_2O$\{$R–COOH + R_1OH \rightleftharpoons R–COOR_1 + H_2O$}$R–COOH + R_1OH \rightleftharpoons R–COOR_1 + H_2O$

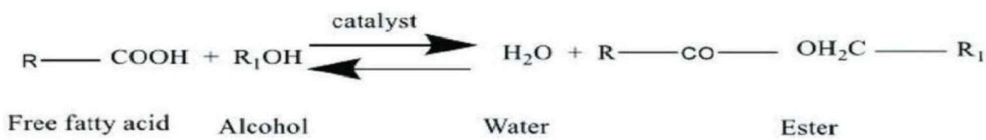


**Figure 4: Pre-Esterification Reaction Mechanism**

The process is executed in three sequential rounds within GST-type agitated reactors operating at **60–70 °C** and **700–800 rpm**, each round employing calibrated methanol and sulfuric-acid dosages to progressively reduce the **acid value (AV)**:

1. **Round 1 :** Oil is charged with 10 % v/v methanol and 0.3 % w/w acid catalyst. Continuous circulation and agitation are maintained until laboratory titration confirms that the AV trend stabilizes ("AV stuck"). The mixture is then allowed to settle for ≈ 1 h, and the aqueous acid phase is drained to slag pools or GB vessels.
2. **Round 2 :** Fresh methanol (7 % v/v) and catalyst (0.2 % w/w) are introduced under identical conditions. Hourly laboratory checks monitor AV until no further decline is observed, after which the system is again settled and the acid water removed.
3. **Round 3 :** A final charge of 5 % v/v methanol and 0.2 % w/w catalyst is added. The reaction proceeds until the target AV of 8–10 mg KOH $g^{-1}$ is achieved. The oil is then settled, acid water drained, and the treated feed transferred to the D405 series tanks for the base-catalyzed trans-esterification stage.

Throughout all rounds, methanol vapor recovery condensers maintain a closed system, and hourly titration data are logged to verify reaction kinetics. The stepwise reduction of FFA not only minimizes soap formation but also ensures chemical congruence between feedstock quality and downstream alkaline-catalyst performance.

In parallel to the main GST reactor system , a dedicated **Glass-Lined Vessel (GLV)** reactor configuration (see **Figure 3 (a)**) is employed for the treatment of recovered or high-acid oils, ensuring uniform feedstock quality across process streams. Distinct from the standard liquid-phase operation, the GLV utilizes a vapor-phase methanol injection mechanism to simultaneously strip residual moisture and methanol (M&M) and promote esterification. Constructed as a seven-shell vessel, each shell equipped with an independent saturated-steam circuit, the GLV maintains internal temperatures of 115–120 °C, while its vaporizer operates between 120–130 °C. Initially, the batch is heated to remove M&M, which is routed through a plate-and-heat exchanger (PHE) to D403 B/A for methanol recovery. Once drying is complete, liquid methanol containing 1 % w/w sulfuric acid catalyst is injected for 10–20 minutes under agitation, followed by sustained vapor-methanol flow at 0.6–0.8 $m^3$ $h^{-1}$. Hourly laboratory sampling monitors the acid value (AV) until stabilization is observed, at which point the system is allowed to settle for 30 minutes, acid water is drained, and vapor injection resumes as required. This cyclical process continues until the target AV of 8–10 mg KOH $g^{-1}$ is achieved, after which liquid-methanol cooling reduces the oil temperature to ≈ 80 °C for safe transfer to D405 reactors for trans-esterification. The integration of the GLV route expands feedstock flexibility, enabling the facility to process low-grade, recovered, or recycled oils without compromising reaction efficiency or downstream catalyst performance.

## 2.4 Trans-Esterification (Methyl-Ester Formation and Initial Separation)

Following the pre-esterification phase, the low-acid oil is transferred to the base-catalyzed trans-esterification system, the principal conversion step where triglycerides react with methanol in the presence of sodium hydroxide (NaOH) to yield FAME—the active biodiesel fraction—and glycerin as a by-product. The overall reaction pathway is illustrated in **Figure 5**, where each triglyceride molecule reacts with three methanol molecules to form a stoichiometric mixture of FAME and glycerin.

Operationally, the process is executed in three plug-flow reactors (PFRs) connected in series to maintain steady-state conditions. Feed oil from D405 tanks is mixed with 17–20 % v/v methanol and 2.7–3 % w/w NaOH prepared in the C401 A/B alkali-mixing units. Reactions are carried out at 60–65 °C under atmospheric pressure, ensuring near-complete triglyceride dissociation. Continuous monitoring through in-line sampling and the standard **"3/27 conversion test"** verifies complete ester formation. The sequential reaction mechanism—from triglyceride to mono- and diglyceride intermediates—is shown schematically in **Figure 5**.

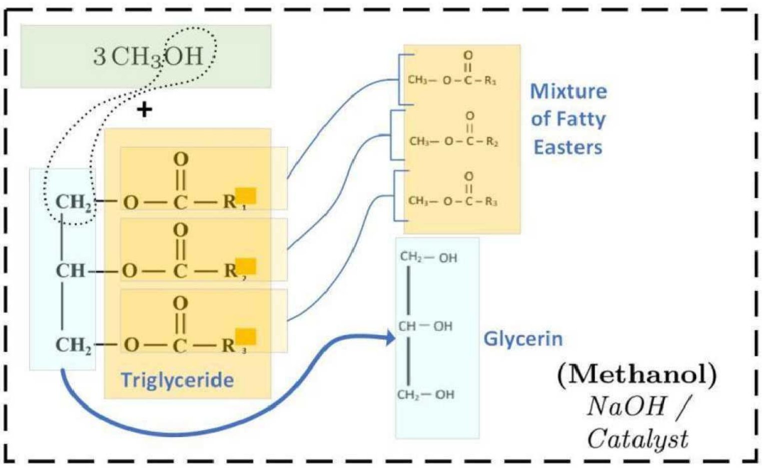


**Figure 5: Fatty acid methyl esterification**

The product mixture exiting the final reactor comprises crude biodiesel, glycerin, unreacted methanol, and trace moisture. It is immediately routed to the falling-film evaporator (E401), where methanol and moisture are stripped under –0.6 bar vacuum before the mixture enters the centrifugal-separation system. The crude biodiesel properties at this intermediate stage (in D402 vessel) are summarized in Table 4, indicating an acid value of 0.2–0.3, soap 0.2–0.5 %, pH 8.5–9, M&M < 4 %, and an oil recovery rate of ≈ 90 %. These values confirm optimal conversion efficiency and acceptable feed quality for subsequent purification and distillation.

## 2.5 Evaporation and Centrifugal Separation

Upon completion of the **trans-esterification reaction**, the reaction effluent—a biphasic mixture of **crude biodiesel, glycerin, methanol, and residual moisture**—is subjected to a two-stage physical purification process comprising methanol removal by falling-film evaporation followed by centrifugal phase separation. This step ensures complete recovery of volatile components and the isolation of high-purity FAME prior to vacuum distillation. In the first stage, the mixture is introduced into the falling-film evaporator (E401), operating under a vacuum of –0.6 bar and a film temperature of 75–80 °C. The thin-film design enhances surface contact and heat transfer, facilitating the rapid removal of M&M without inducing thermal degradation. Vapors are condensed through a sequence of spiral condensers (E404, E402) and collected in D404, where recovered methanol typically exceeds 98 % purity and is recycled upstream to both esterification sections. This closed-loop configuration significantly reduces solvent losses and emissions, aligning the process with zero-liquid-discharge (ZLD) objectives. The process parameters of the centrifuge stage are shown in **Figure 6** and **Table 2**.


**Figure 6: Centrifuge operational parameters**

**Table 2: Centrifuge Operating Parameters during Hot-Water and Feed Cycles**

| | **On Hot Water** | **On Feed** |
|---|---|---|
| Vibration Intensity | 0.3-0.5mm/s | 1.0-1.5mm/s |
| Ampere | 18-20A | 20-25A |
| RPM | 65-70 | 65 |
| Partial discharge time | 300sec | 300sec |

The degassed oil–glycerin mixture from E401 is then conveyed to the centrifugal-separation unit, comprising two high-speed centrifuges arranged in series. At an inlet temperature of ≈ 70 °C, the first centrifuge (CF-1) separates the lighter biodiesel phase, while the heavier soap and glycerin phase is directed to CF-2 for secondary clarification. Optimal operation is maintained by regulating rotor speed ≈ 65 Hz, current draw 20–25 A, and vibration amplitude 1.0–1.5 mm $s^{-1}$; deviations trigger automatic system alarms and controlled shutdowns for plate inspection. A hot-water flow of 0.2–0.4 $m^3$ $s^{-1}$ prevents viscosity buildup and maintains continuous discharge.

The separated crude biodiesel—collected in D402—is characterized by stable physicochemical properties: acid value 0.2–0.3, soap 0.2–0.5 %, pH 8.5–9, M&M < 4 %, and an oil recovery rate of ≈ 90 %. These metrics validate efficient phase separation and prepare the product for subsequent vacuum-distillation refining, where residual impurities are polished to meet EN 14214 standards.

## 2.6 Vacuum Distillation and Purification

Following centrifugal clarification, the crude biodiesel collected in D402 is subjected to vacuum distillation for final purification. This process removes trace contaminants such as residual methanol, moisture, light diesel oil (LDO), and sulfur species, producing fuel that meets EN 14214 and ASTM D 6751 standards. The distillation system comprises a multi-stage vacuum network and dual packed-tower configuration (T501 and T502) operating under high-vacuum and moderate-temperature conditions. The complete process layout is shown in **Figure 7 (a)**, while the internal tower tray and packing configurations are illustrated in **Figure 7 (b)**.

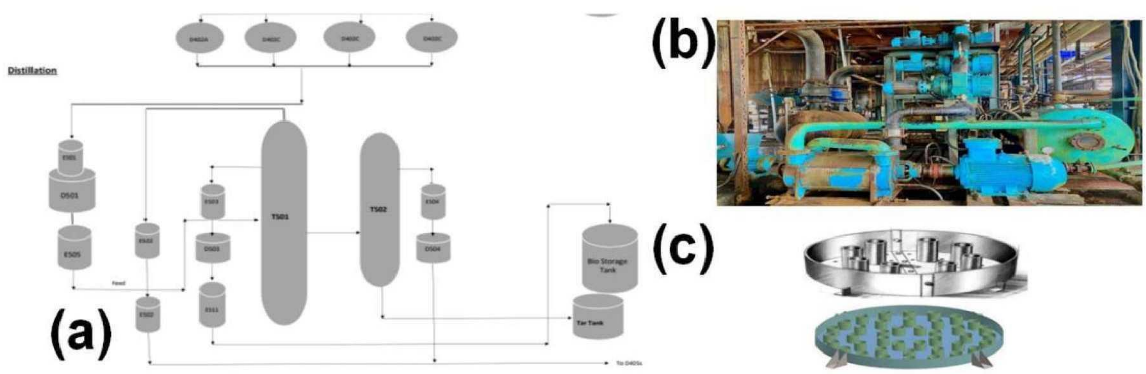


**Figure 7: (a) Vacuum Distillation and Purification System Flow Diagram; (b) Packed-Tower Internal Configurations; (c) Vacuum Pump Assembly**

### 2.6.1 System Configuration and Operating Principle

The feed from D402 first enters the M&M removal column (D501), where it is heated to approximately 180 °C under a vacuum of –0.5 to –0.7 bar. This operation lowers the moisture content to < 0.3 %, as summarized in **Table 3**. The pre-treated stream then receives a 0.1 % nickel solution at 150 °C, enabling sulfur adsorption to below 20 ppm, before being pumped by E505 to the distillation towers.

**Table 3: D501 Operating Parameters**

| Name | Parameters to follow |
|---|---|
| D501 Temp° | 180°C |
| D501 Vacuum | -0.5 to -0.7bar |

| | |
|---|---|
| D501 Out Moisture | <0.3% |
| D501 Level | 40% |
| Buffer Tank level | No level |

The primary tower (T501) functions under a deep vacuum of 2–3 torr, achieved by a four-stage vacuum system consisting of RVPA–RVPD pumps that progressively reduce pressure from 48 torr to < 1 torr (see Table 4). The column operates with a feed flow of 4–5 MT $h^{-1}$, a bottom temperature of 230–240 °C, and a top temperature exceeding 75 °C, ensuring effective vapor–liquid separation. Vapors rich in light hydrocarbons and methanol are condensed through E503 and E511, while the refined FAME distillate is collected in the Bio Storage Tank at 30–35 °C.

**Table 4: Vacuum Network Staging and Pressure Levels**

| Pump | Vacuum Created |
|---|---|
| RVPA | 48 torr (from 760 torr) |
| RVPB | 18 torr |
| RVPC | 4 torr |
| RVPD | < 1 torr |

A controlled reboiler reflux ratio maintains temperature uniformity across the packed beds. The secondary tower (T502) receives a portion of the heavy stream from T501 to recover additional biodiesel fractions, operating at a bottom temperature of ~260 °C. The heavier residues and non-volatile tar components are bled into a Tar Tank, preventing column fouling and sustaining operational stability. The flow pattern between the two towers and auxiliary heat exchangers is presented in **Figure 6**.

### 2.6.2 Process Performance and Product Quality

The final biodiesel product exhibits superior clarity, reduced acid value, and improved flash point, rendering it suitable for blending and aviation-grade upgrading. Table 6c compares product quality between acid-oil (AO) and recovered-oil (RO) feedstocks, demonstrating equivalent process fidelity with minor variation in yield. The AO stream consistently achieves 90 %+ oil rate at acid value 0.20, while RO batches record 85 % yield at acid value 0.30, both maintaining soap < 0.5 % and M&M < 4 %.

**Table 5: Product Quality and Feed Comparison for AO and RO Batches.**

| Parameter | Range |
|---|---|
| Feed Flow | 4-5MT/hr. |
| Vacuum | 2-3torr |
| T501 level | 20% (10″) |
| D503 Level | 10% (10″) |
| Column1Bottom Temperature | 230-240 $^0$C |
| Column1Mid Temperature | >200 $^0$C |
| Column1 Top Temperature | >75 $^0$C |
| Column2 Bottom Temperature | 260 $^0$C |

## 2.7 By-Product Recovery and Circular Integration

A defining characteristic of the facility is its closed-loop recovery system, which converts all secondary streams—soap, recovered oil, and methanol vapors—into reusable intermediates. This integration strengthens material efficiency, reduces waste, and aligns the operation with ZLD principles. The overall configuration of the by-product and solvent-recovery network is presented in **Figure 8**.

### 2.7.1 Soap Formation and Conversion Mechanism

During trans-esterification, unavoidable contact between residual **free-fatty acids (FFAs)** and the alkaline catalyst (NaOH) results in saponification, forming soap as a by-product. The reaction sequence (Equation 1) occurs through hydrolysis of triglycerides into FFAs and diglycerides, followed by neutralization of FFAs with sodium hydroxide:

$$\begin{aligned} &Triglyceride + H2 \quad O \rightleftharpoons Diglyceride + R\text{–}COOHR\text{–}COOH + NaOH \\ &\quad \rightarrow R\text{–}COONa + H2OR\text{–}COOH + NaOH \\ &\quad \rightarrow R\text{–}COONa + H_2OR\text{–}COOH + NaOH \rightarrow R\text{–}COONa + H2 \end{aligned}$$

*(Equation 2.7.1 — Soap-Formation Reactions)*

The soap phase, discharged via Nozzle 3 of the centrifuge system, is directed to the Fatty-Residue Processing (FRP) line for recovery rather than disposal.

### 2.7.2 Recovered-Oil Formation (Acidulation Process)

Within the FRP reactors, the separated soap is chemically acidulated using hydrochloric acid (HCl) to regenerate free fatty acids (R–COOH) according to Equation 2:

$$\begin{aligned} &R\text{–}COONa + HCl \rightarrow R\text{–}COOH + NaClR\text{–}COONa + HCl \\ &\quad \rightarrow R\text{–}COOH + NaClR\text{–}COONa + HCl \rightarrow R\text{–}COOH + NaCl \end{aligned}$$

*(Equation 2.7.2 — Recovered-Oil Formation Reaction)*

The acidulated mixture is processed through a tertiary centrifuge (CF-3) to remove entrained water, residual methanol, and solids. The purified recovered-oil stream is reintroduced into the GLV pre-esterification units, providing a renewable feed source that enhances plant circularity and minimizes organic loss. Operating parameters and yield efficiencies for this stage are summarized in **Table 6**.

### 2.7.3 Methanol-Recovery and Reuse Loop

A fully integrated methanol-recovery train captures vapors from multiple points—E401 (falling-film evaporator), D203, D403 A/B, and D404—and channels them through a packed column (T601) operating under controlled vacuum. The column separates entrained oil and water by gravity, while purified methanol vapors are condensed in spiral condensers E404 and E402. Condensed methanol of ≥ 98 % purity is collected in D404 and transferred via a buffer tank to the methanol-storage system for reuse in both pre- and trans-esterification reactors. Uncondensed vapors pass through an inclined shell-and-tube exchanger for secondary condensation, ensuring near-total solvent recovery. The schematic of this loop is shown in **Figure 8**: showing its automated control visualization with temperature, flow, and vacuum readouts. Key operating conditions—temperatures, vacuum range, and solvent-purity results—are detailed in **Table 6**.

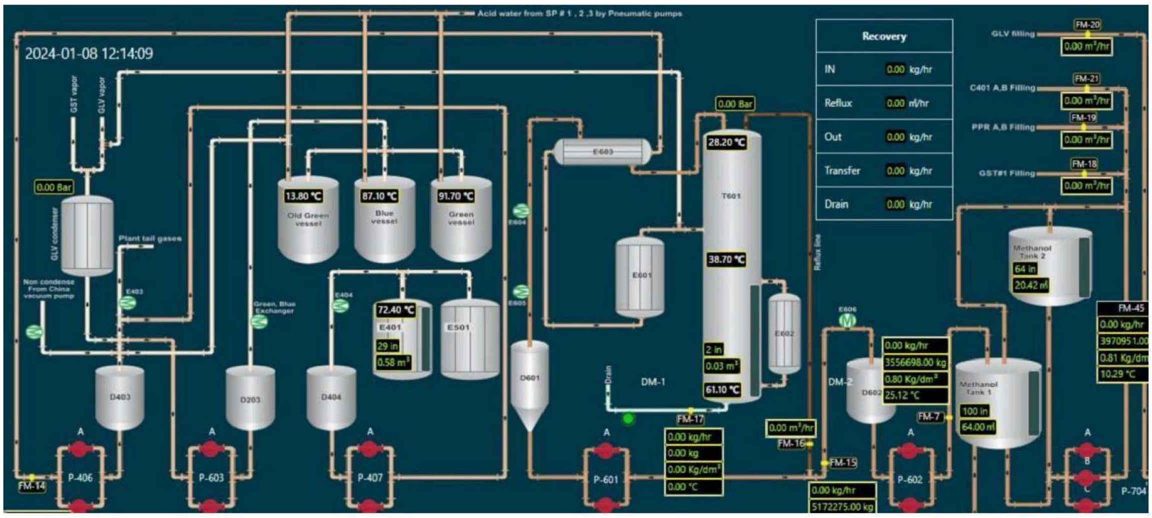


**Figure 8: Process-Control Interface of the Methanol-Recovery System**

Together, these interlinked recovery systems transform potential waste into valuable inputs, ensuring process sustainability, economic resilience, and environmental compliance for Pakistan's first integrated bio-jet-fuel production facility.

**Table 6: Operational Parameters and Product Quality Data for Soap, Recovered-Oil, and Methanol-Recovery Units**

| Subsystem | Key Parameters | Operating Range / Typical Value | Performance Indicator |
|---|---|---|---|
| Soap Formation (Centrifuge Nozzle 3) | Discharge temperature | 65–70 °C | Stable separation, no emulsion carryover |
| | Soap flow rate | 0.4–0.6 $m^3\ h^{-1}$ | Consistent stream for FRP feed |
| Recovered-Oil Formation (FRP / CF-3) | Acid concentration (HCl) | 5–7 % v/v | Complete neutralization of soap |
| | Reaction temperature | 95–105 °C | Efficient acidulation without degradation |
| | Settling/centrifugation time | 20–30 min | ≥ 95 % phase separation |

| Subsystem | Key Parameters | Operating Range / Typical Value | Performance Indicator |
|---|---|---|---|
| | Final oil acid value | 8–10 mg KOH $g^{-1}$ | Ready for GLV pre-esterification |
| | Oil yield (recovery rate) | 85–90 % | Minimal organic loss |
| | Operating temperature (E401) | 70–80 °C | Optimal vaporization rate |
| | Column vacuum (T601) | –0.4 to –0.6 bar | Stable solvent separation |
| Methanol-Recovery Loop (E401–E404–E402–D404–T601) | Condenser outlet temperature | 25–30 °C | Effective condensation |
| | Methanol purity | ≥ 98 % (w/w) | Verified by laboratory GC analysis |
| | Recycle rate to process | 95–98 % | Achieves closed solvent balance |

# 3 Results and Applications

## 3.1 Overall Process Performance

The integrated biodiesel and bio-jet fuel demonstration plant operated by **Bio Tech Energy (Pvt.) Ltd.**, under **SAFCO Ventures**, in Sheikhupura, Pakistan, represents the nation's first full-scale biofuel production system utilizing waste-derived feedstocks. The process combines acid- and base-catalyzed esterification, vacuum distillation, and closed-loop methanol recovery within a continuous, digitally monitored configuration.

Average performance evaluation over multiple operational runs confirms a conversion efficiency exceeding 98 %, with final acid values ≤ 0.30 mg KOH $g^{-1}$, and soap content < 0.5 %. Biodiesel yields reach 90 % ± 2 % for acid-oil feedstocks and 85 % ± 3 % for recovered-oil batches, indicating strong congruence between feedstock variability and reaction robustness.

The plant's solvent-recovery network attains a methanol-purity ≥ 98 % and a recycle rate of 95–98 %, substantially reducing operating expenditure and environmental load. The distillation unit achieves sulfur reduction below 20 ppm and ensures compliance with EN 14214 biodiesel and ASTM D 1655 jet-fuel specifications.

**Table 7: Operational Results and Performance Indicators**

| Parameter | Unit / Basis | Measured / Calculated Value | Performance Benchmark | Remarks / Inference |
|---|---|---|---|---|
| Feedstock throughput (total) | t $yr^{-1}$ | ≈ 45,000 (t Biodiesel Plant); 200,000 planned (SAF Facility) | — | Existing + Expansion capacity (IFC / ADB data). |
| Feedstock composition | — | UCO 40 %, Acid Oil 30 %, Animal Fats 20 %, Others 10 % | — | Typical waste-oil mix for Pakistan region. |
| Moisture after dehydration | % w/w | 0.2–0.3 | ≤ 0.3 | Achieved in D101/E101 stage. |
| Pre-esterification conversion (FFA → FAME) | % | ≈ 95 | ≥ 90 | Three-round acid-catalyzed process. |
| Trans-esterification conversion | % | > 98 | ≥ 96 | Validated via 3/27 test. |
| Final acid value (AV) | mg KOH $g^{-1}$ | 0.20 – 0.30 | ≤ 0.50 | Conforms to EN 14214. |
| Soap content | % w/w | 0.2 – 0.5 | < 0.5 | Within limits after centrifugation. |
| Moisture & Methanol (M&M) | % | < 4.0 | < 5 | Controlled via E401 evaporator. |
| Sulfur content (after Ni polish) | ppm | < 20 | ≤ 20 | Meets aviation grade requirement. |
| Biodiesel yield (AO feed) | % by mass | ≈ 90 | > 85 | Consistent with acid-oil batches. |
| Biodiesel yield (RO feed) | % by mass | ≈ 85 | > 80 | Recovered oil processing efficiency. |
| Methanol recovery efficiency | % | 95–98 | ≥ 95 | High solvent recycle rate. |
| Methanol purity (D404/D602) | % w/w | 98–99 | ≥ 98 | GC-confirmed purity. |
| Energy consumption (process avg.) | kWh $t^{-1}$ biofuel | ≈ 220 – 250 | ≤ 300 | Includes steam and vacuum loads. |
| GHG reduction vs. fossil diesel | % ( LCA basis ) | ≈ 75–80 | > 65 | Based on IFC/ADB impact model. |
| Product flash point | °C | > 130 | > 120 | Meets EN 14214 spec. |
| Kinematic viscosity (40 °C) | $mm^2\ s^{-1}$ | 4.3–4.7 | 3.5–5.0 | Within range. |
| Density (15 °C) | kg $m^{-3}$ | ≈ 875 | 860–900 | Typical for FAME. |
| Cold filter plugging point (CFPP) | °C | –1 to +3 | ≤ +5 | Favorable for temperate operations. |
| Distillation residue (tar) | % | < 2.0 | < 3 | Stable tower operation. |

## 3.2 Application and Industrial Impact

The integrated system demonstrates industrial viability and environmental compatibility for Pakistan's emerging green-fuel sector. By sourcing waste-based feedstocks from urban centers such as Lahore, Gujranwala, and Faisalabad, the facility converts low-value residues into high-value clean fuel while supporting local waste-management frameworks.

The project's expansion—supported by **IFC (USD 35 million)** and **ADB (USD 86.2 million)**—is projected to increase capacity to **200,000 t $yr^{-1}$ of SAF** and **30,000 t $yr^{-1}$ of bio-naphtha**, reducing Pakistan's aviation $CO_2$ footprint by ≈ **500,000 t $CO_2$-e annually**. Integration with **Axens VEGA and Esterfip technologies** (under evaluation) further broadens conversion flexibility and ensures compatibility with ASTM D 7566 aviation-fuel standards.

From an economic perspective, high solvent-recycle efficiency and feedstock circularity reduce OPEX by an estimated 12–15 %, while the sale of glycerin (by-product) and recovered oil offsets raw-material costs. Socially, the facility supports regional employment in feedstock logistics and plant operations, fostering rural-urban value-chain linkages.

## 3.3 Product Quality and Standards Compliance

Comprehensive laboratory analysis confirms that the biodiesel and bio-jet-fuel fractions produced through the described integrated process comply with all key international fuel-quality specifications. The final FAME product meets the physicochemical limits defined in the European Standard EN 14214 [21] for biodiesel, including constraints on acid value, sulfur, total glycerin, and oxidative stability. Likewise, distillate fractions refined under deep-vacuum operation conform to ASTM D 6751 [22] for biodiesel and ASTM D 1655 [23] for aviation-grade kerosene, ensuring compatibility with downstream upgrading to SAF. Independent verification of parameters such as flash point (> 130 °C), density (≈ 875 kg $m^{-3}$), viscosity (4.3–4.7 $mm^2\ s^{-1}$ at 40 °C), and sulfur (< 20 ppm) demonstrates full alignment with these standards. These findings confirm that the process yields a stable, high-purity renewable fuel suitable for both domestic blending programs and export markets compliant with global sustainability criteria.

# 4 Conclusion

The establishment of Pakistan's first integrated biofuel and bio-jet-fuel production facility marks a transformative milestone in the nation's transition toward sustainable energy. By demonstrating the industrial viability of converting waste-derived feedstocks—including used cooking oils, poultry fats, and acid oils—into high-grade FAME and distillate fuels, the project provides a functional model for circular resource utilization within the energy sector. The process's closed-loop design—encompassing feedstock pre-treatment, staged esterification, methanol recovery, and by-product recycling—ensures near-zero waste generation and minimal environmental discharge.

The successful implementation of this system validates that **renewable-fuel production from domestic organic residues** can be both technologically robust and economically feasible. Beyond direct emission reductions, the plant's integration of solvent recovery and feedstock re-use embodies the broader principles of resource efficiency and climate resilience emphasized in national and international sustainability frameworks. The initiative thereby establishes a foundation upon which Pakistan can expand its renewable-energy portfolio, reduce dependence on imported fossil fuels, and align with global low-carbon aviation strategies.

More broadly, the results underscore that targeted industrial innovation, policy alignment, and scientific collaboration can bridge the gap between laboratory-scale feasibility and full-scale green-energy deployment in developing economies. The demonstrated process serves not only as an operational success but as a template for replication across South Asia—linking agricultural by-products, municipal waste management, and clean-fuel production into a single, coherent pathway toward a greener future.

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Special recognition is also given to the **plant engineering and process-operations teams** for their technical collaboration and laboratory assistance during data verification and process monitoring. Their contributions have been instrumental in translating conceptual process design into a fully operational industrial system that now stands as a benchmark for sustainable fuel innovation in Pakistan.

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